

**NIRMALA MEMORIAL FOUNDATION COLLEGE OF COMMERCE AND SCIENCE
(AUTONOMOUS)**

Re-Accredited by NAAC with B++ Grade & ISO 9001:2015 Certified Recognized under Section 2 (f) & 12 (B) of the UGC Act, 1956.
Kandivali (East), Mumbai-400101.



**IoT-Based Bus Stop System for Public Transport Overcrowding Prediction
Using Machine Learning**

A Project Report

Submitted in partial fulfillment of the requirements For the degree of B.Sc. Information Technology aligned with
the NEP-2020 Community Engagement Project (CEP)

By

Mr. Om Keluskar and Mr. Jashith Agre

Seat No: 24TIT078, 24TIT011

Semester: V

Under the esteemed guidance of

Mr. George Thekkevilayil
Assistant Professor

BACHELOR OF SCIENCE INFORMATION TECHNOLOGY

Academic Year: 2026–2027



Nirmala Memorial Foundation College of Commerce and Science

(Autonomous)

Re-Accredited by NAAC with B++ Grade & ISO 9001:2015 Certified
Recognized under Section 2 (f) & 12 (B) of the UGC Act, 1956

D S Road, Asha Nagar, Thakur Complex, Kandivali (East), Mumbai-400101 Tel: 022-69436400

CERTIFICATE OF PROJECT COMPLETION

B.Sc. (Information Technology)

This is to certify that Mr. Om Keluskar and Mr. Jashith Agre, students of T.Y.B.Sc. (Information Technology), Semester V, bearing Seat No. 24TIT078 and 24TIT011, has successfully completed the Project Work titled "IoT-Based Bus Stop System for Public Transport Overcrowding Prediction Using Machine Learning." The project has been submitted in partial fulfillment of the requirements prescribed in the syllabus for the award of the Degree of Bachelor of Science in Information Technology for the academic year 2026–2027. The practical and project work has been carried out successfully under the supervision of the undersigned.

Internal Examiner

External Examiner

Co-ordinator

Prof. George Thekkevilayil

Prof. Vaishali Mishra

College seal and Date



Nirmala Memorial Foundation College of Commerce and Science (Autonomous)

Re-Accredited by NAAC with B++ Grade & ISO 9001:2015 Certified
Recognized under Section 2 (f) & 12 (B) of the UGC Act, 1956

D S Road, Asha Nagar, Thakur Complex, Kandivali (East), Mumbai-400101 Tel:022-69436400

PROJECT PROPOSAL APPROVAL SHEET

**Department of Information Technology
Academic Year 2026-27**

1. Name of the Students: Mr. Jashith Agre and Mr. Om Mangesh Keluskar

2. Seat Number: 24TIT011, 24TIT078

3. Is this your first submission? Yes ☐ No ☐

4. Name of the Guide: Prof. George Thekkevilayil

5. Teaching Experience of the Guide: 26 Years

The proposed project titled "**Iot-Based Bus Stop System for Public Transport Overcrowding Prediction Using Machine Learning.**" submitted by the above students, has been examined and found suitable for the partial fulfilment of the requirements for **Semester V** of the B.Sc IT Degree Programme. The project is hereby approved for implementation under the guidance of the undersigned.

Signature of Guide

Date:

Signature of Co-ordinator

Date:

TABLE OF CONTENTS

CHAPTER 1: INTRODUCTION AND COMMUNITY CONTEXT	1
1.1. Project Background	1
1.2. Community Problem Identification	1
1.3. Community Need Assessment	2
1.4. Existing System Analysis	3
1.5. Problem Statement	3
1.6. Proposed Solution	3
1.7. Aim and Objectives	4
1.8. Scope of Project	4
1.9. Limitations	5
1.10. Major Modules	5
1.11. Success Criteria	6
1.12. Expected Outcomes	6
1.13. Chapter Summary	7
CHAPTER 2: SYSTEM ANALYSIS AND REQUIREMENT ANALYSIS	8
2.1. Introduction	8
2.2. Literature Review	8
2.3. Comparative Analysis	11
2.4. Research Gap Identification	11
2.5. Requirement Gathering	12
2.6. Stakeholder Analysis	13
2.7. Functional Requirements	14
2.8. Nonfunctional Requirements	15
2.9. Feasibility Study	16
2.10. Software Requirements	16
2.11. Hardware Requirements	17
2.12. Project Schedule	17
2.13. Development Methodology	18

CHAPTER 3: SYSTEM DESIGN	19
3.1. System Architecture	19
3.2. System Workflow	20
3.3. Block Diagram	21
3.4. Functional Flow Diagram	22
3.5. Circuit Diagram	23
3.6. Component Interconnection Diagram	25
3.7. Hardware Module Design	27
3.8. Data Flow & Processing Logic	30
3.9. Database Design (if applicable)	31
3.10. Dashboard/UI Design	32
3.11. Technology Stack	33
3.12. Safety & Security Considerations	34
3.13. Chapter Summary	35
CHAPTER 4: DEVELOPMENT AND IMPLEMENTATION	36
4.1. Development Environment	36
4.2. Implementation Approach	36
4.3. Hardware Assembly	38
4.4. Firmware Development	42
4.5. Database/Cloud Integration	44
4.6. Dashboard Development	46
4.7. Module-wise Implementation	48
4.8. Algorithms and Control Logic	50
4.9. Safety Features Implemented	52
4.10. Prototype Development	53
4.11. Prototype Photos & Demonstration	54

CHAPTER 5: TESTING, VALIDATION AND PERFORMANCE EVALUATION	58
5.1. Testing Strategy	58
5.2. Unit Testing	58
5.3. Integration Testing	60
5.4. System Testing	62
5.5. User Acceptance Testing	64
5.6. Performance Evaluation	65
5.7. Security Testing	67
5.8. Validation Metrics	68
5.9. Test Cases	69
5.10. Defects Identified and corrections	70
5.11. Chapter Summary	71
CHAPTER 6: COMMUNITY DEPLOYMENT AND IMPACT ASSESSMENT	72
6.1. Beneficiary Profile	72
6.2. Community Engagement Activities	72
6.3. Pilot Deployment	73
6.4. User Training	74
6.5. Feedback Collection	75
6.6. Feedback Analysis	75
6.7. Impact Assessment	77
6.8. Outcome Achievement Matrix	78
6.9. Sustainability Plan	79
CHAPTER 7: RESULTS, DISCUSSION, CONCLUSION AND FUTURE SCOPE	80
7.1. Results and Discussion	80
7.2. Achievement of Objectives	81
7.3. Project Contributions	82
7.4. Limitations	83
7.5. Future Enhancements	84
7.6. Conclusion	85
REFERENCES	86
ANNEXURES	89 onwards



**Nirmala Memorial Foundation College of
Commerce and Science
(Autonomous)**

**Department of Information Technology
Community Engagement Project Questionnaire**

Name of Student(s): Mr.Jashith Agre, Mr.Om Mangesh Keluskar

Seat No: 24TIT011,24TIT078

Name of Participant:

Signature of Participant

Survey Questionnaire

1. How frequently do you experience overcrowding at Dindoshi Bus Depot or the BEST Route 326 bus stop? ☐ Every Day ☐ Several Times a Week ☐ Occasionally ☐ Rarely
2. Which mode of transport do you use most frequently to reach this stop? ☐ BEST bus (Route 326) ☐ Other BEST Bus ☐ Train then BEST Bus ☐ Walking
3. What is the biggest problem you face because of overcrowding at the bus stop? ☐ Missed bus / cannot board ☐ Pushing or injury risk ☐ Long uncomfortable wait ☐ No crowding warning
4. Have you ever been unable to board a bus because the stop or doorway was too crowded? ☐ Yes, Multiple Times ☐ Yes, Once ☐ No ☐ Not Applicable
5. How safe do you feel while waiting at an overcrowded bus stop? ☐ Very Safe ☐ Safe ☐ Unsafe ☐ Very Unsafe
6. Have you ever reported overcrowding to depot staff or BEST officials? ☐ Yes ☐ No ☐ Didn't Know How ☐ Never Felt It Would Help
7. If you did report overcrowding, how satisfied were you with the response? ☐ Very Satisfied ☐ Satisfied ☐ Dissatisfied ☐ Never Reported
8. How important is it to warn commuters as soon as overcrowding is detected? ☐ Extremely Important ☐ Important ☐ Slightly Important ☐ Not Important
9. Do you think rain, heat, or peak hours make overcrowding at the stop more dangerous? ☐ Strongly Agree ☐ Agree ☐ Disagree ☐ Strongly Disagree
10. Would you trust an automated IoT system that predicts overcrowding, without requiring a mobile app? ☐ Definitely Yes ☐ Probably Yes ☐ Probably No ☐ Definitely No
11. Which feature would be most useful in a smart bus-stop overcrowding system? ☐ On-site LED display ☐ Telegram staff alerts ☐ Live risk percentage ☐ Bus-left operator button
12. Do you believe an IoT-based pothole detection system can improve road safety and municipal maintenance? ☐ Strongly Agree ☐ Agree ☐ Disagree ☐ Strongly Disagree



**Nirmala Memorial Foundation College of
Commerce and Science
(Autonomous)**

**Department of Information Technology
Community Engagement Project Questionnaire**

Name of Student(s): Mr.Jashith Agre, Mr.Om Mangesh Keluskar

Seat No: 24TIT011,24TIT078

Name of Participant:

Signature of Participant

Feedback Questionnaire

1. How easy was it to understand the purpose of the IoT bus-stop overcrowding system?

☐ VeryEasy ☐ Easy ☐ Neutral ☐ Difficult

2.How useful do you think automatic passenger counting using the IoT sensor would be for overcrowding awareness?

☐ VeryUseful ☐ Useful ☐ ModeratelyUseful ☐ Not Useful at All

3.How clear and understandable was the demonstrated overcrowding-prediction process?

☐ VeryClear ☐ Clear ☐ Neutral ☐ Unclear

4.How useful were the on-site red and green LED indicators for understanding crowding?

☐ Useful ☐ ModeratelyUseful ☐ SlightlyUseful ☐ Not Useful at All

5.How useful would real-time Telegram alerts be for depot staff when crowding becomes CRITICAL?

☐ VeryEasy ☐ Easy ☐ Neutral ☐ Difficult

6.How easy was it to understand the crowding information shown on the dashboard (count, weather, risk, status)?

☐ VeryEasy ☐ Easy ☐ Neutral ☐ VeryDifficult

7.How clearly did the hardware demonstration show how the sensor detects passenger crossings?

☐ VeryClear ☐ Clear ☐ Neutral ☐ Unclear

8.How confident are you that this system can warn of overcrowding faster than manual visual guessing?

☐ VeryConfident ☐ Confident ☐ ModeratelyConfident ☐ SlightlyConfident

9.Would you be willing to use this system if it remained available at Dindoshi Route 326?

☐ DefinitelyYes ☐ ProbablyYes ☐ Not Sure ☐ DefinitelyNo

10.How clear and easy to understand were the NORMAL, WARNING, and CRITICAL status lights?

☐ VeryClear ☐ Clear ☐ Neutral ☐ Unclear

11.How useful do you think this system would be for improving crowding awareness at your bus stop?

☐ VeryUseful ☐ Useful ☐ ModeratelyUseful ☐ Not Useful at All

12.Overall, how satisfied are you with the IoT bus-stop overcrowding system and its potential for commuter safety?

☐ VerySatisfied ☐ Satisfied ☐ Neutral ☐ Dissatisfied